



Department of Electronics and Communication Engineering
Academic Year 2021 - 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE-A

Course Code & Title: EC8791 Embedded and Real Time Systems

Name of the Faculty member: Mr.B.Kannan, AP (SG)

Innovative Practice Description

- **Unit/Topic:** Unit-III / Software Performance Optimization
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen:** Flipped Class
- **Justification:**

A flipped classroom (also called a reverse, inverse, or backward classroom) is a pedagogical approach in which basic concepts are provided to students for pre-class learning so that class time can apply and build upon those basic concepts. The topic chosen to perform the activity is software performance optimization. It includes sub-topics that elaborately discuss the loop reduction in the program and the utilization of cache memory in a better way to optimize the performance of the software.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The materials for the preparation of the students were shared one week before through LMS Canvas. Including the material, students were asked to refer to the web content also.

(ii) Formation of Groups:

As per the number of sub-topics, student groups were created as below:

- Number of sub-topics: 06
- Class Strength: 36
- Number of groups: 06
- Members per group: 06

(iii) Topic Allocation:

For six sub-topics, students were divided into six groups, and topics were allotted in advance. The students must prepare a 5-minute presentation, including the topic learned, 3 minutes for questions and answers, and a description of each member's participation.

(iv) Topic Delivery:

The main objective of the flipped classroom activity is that each group will learn one application, and the presentation helps the other members learn about the topic as well.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PO10	PSO2
CO3	2	2	1	1	2

• **PO / PSO mapped:**

Innovative Practice	PO9	PO10	PSO2
	1	1	3
Justification for correlation	The group assignment ensures the submission of a technical report and recognizes a variety of working and learning preferences as well as appreciates the value of diversity in a team. Thus, the course outcome is mapped at level 1.	The, course outcome is mapped at level 1. The assignment enables the students to interpret various case studies to model embedded systems using the ARM processor, and it will be available as a document for reference.	The latest concepts in programming help to satisfy the end-user. Hence, the course outcome is mapped at level 3.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ *Feedback on practice from students and other stakeholders:*

- The material shared with us helped a lot with slide preparation.
- The choice of group formation given to the students provided an opportunity to pick their team members.

❖ *Benefits of the practice:*

- Because of a creative approach rather than traditional practice, outcome achievement might have improved.

- Many students presented their slides with some examples without any stage fright.
- The activity helped the students do their roll better.
- The exercise is designed for the students, and it includes interactions and discussions that help students better understand the topics.
- ❖ ***Challenges faced in implementation:***
 - Time management is a big challenge because few students delivered their topics elaborately.
 - Not able to get a few students participation levels up to the expected level due to stage fear.
 - Among six groups, the groups took extra time to complete their presentations, so two groups were not able to utilize their time fully for the presentation.
 - Need to allot more time for question and answer sessions.

References:

- ❖ Marilyn Wolf, "Computers as Components - Principles of Embedded Computing System Design", Third Edition "Morgan Kaufmann Publisher (An imprint from Elsevier), 2012.
- ❖ <https://www.edutopia.org/article/4-tools-flipped-classroom>.
- ❖ David. E. Simon, "An Embedded Software Primer", 1st Edition, Fifth Impression, Addison-Wesley Professional, 2007.
- ❖ <https://medium.com/the-andela-way/performance-optimization-in-software-development-ae7952ab885e>.
- ❖ <https://www.mercuryworks.com/blog/optimize-software-performance/>



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Innovative Practice Description

- **Unit/Topic:** Unit-IV / Fault Tolerance Techniques
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Fraternize Group
- **Justification:**

Fault tolerance is the process of operating a system properly despite the occurrence of failures in the system. Even after performing so many testing processes, there is the possibility of failure in the system. Practically, a system can't be made entirely error-free. Hence, students will understand the system design with fault-tolerance to work properly and give the correct result. This fraternize group activity is intended to help students remember what they've learned and improve their learning through teamwork.

- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

(i) Materials for the Activity:

The topic for the groups was distributed at the time of the activity's proceeding. I informed them of the activity date two days before and asked the students to bring a hard copy of the material for their preparation. Students were also required to refer to the web content in addition to the material.

(ii) Formation of Groups:

As per the number of sub-topics, student groups were created as below:

- Number of Topics: 06
- Class Strength: 30
- Number of groups: 06
- Members per group: 05

(iii) Topic Allocation:

For six sub-topics, students were divided into six groups, and topics were allotted on the spot. The student's group discussed the topic in detail for 20 minutes with the materials.

(iv) Topic Delivery:

From each group, one person joined the other group to disseminate the topics learned. The students have to answer questions from other group members.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO9	PSO2
CO3	3	2	1	2

• **PO / PSO mapped:**

Innovative practice	PO9	PSO2
	1	3
Justification for correlation	As team students prepare a topic for the innovative activity, it enables the students to learn and diverse the content as an individual and as a team. Thus, the course outcome is mapped at level 1.	The best real-time system helps to design an embedded system to satisfy the end-user. Hence, the course outcome is mapped at level 3.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback on practice from students and other stakeholders:**

- This activity helps to share the topics simultaneously among all groups.
- They believed that this form of exercise boosted their conscientious abilities.
- It aids in the development of effective communication skills and active learning.

❖ **Benefits of the practice:**

- It created an opportunity for the students to deliver their topics to other members of the group.
- Knowledge sharing on different topics within 50 minutes is possible through this activity.
- All students actively participated without hesitation.

❖ **Challenges faced in implementation:**

- A few students hesitated to express the learned topics because of the group shifting from parent groups to other groups.
- Extra time was taken to rearrange themselves to form a new group.
- A few groups asked for extra time for preparation.

References:

- ❖ Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan Kaufmann Publisher (An imprint from Elsevier), 2012.
- ❖ David. E. Simon, “An Embedded Software Primer”, 1st Edition, Fifth Impression, Addison-Wesley Professional, 2007.
- ❖ <https://www.amle.org/innovative-teaching-strategies-that-improve-student-engagement/>.
- ❖ <https://www.edsys.in/16-innovative-ideas-make-teaching-methods-effective/>



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Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (Odd Semester)

Active Learning Description for Unit-I

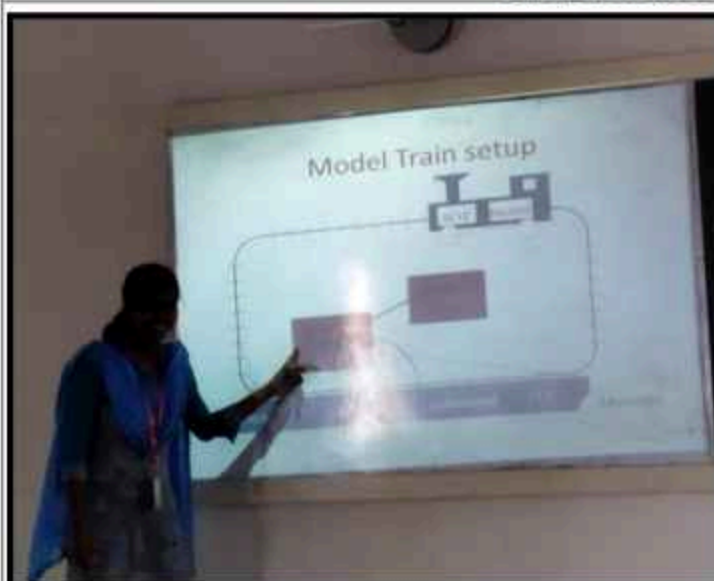
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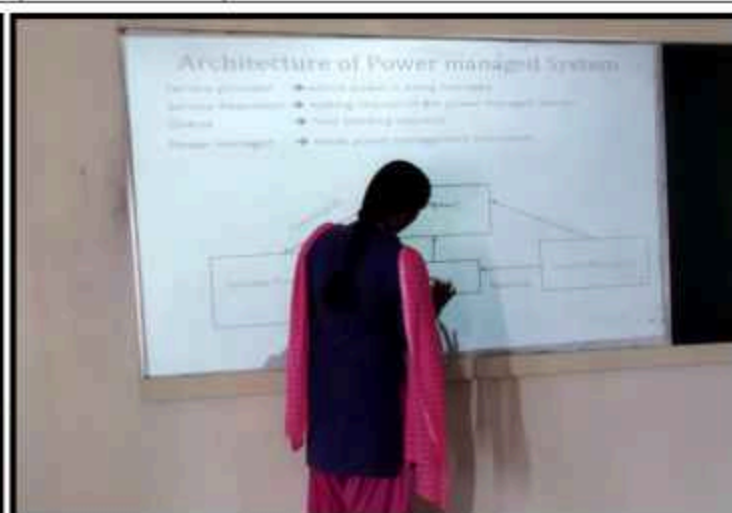
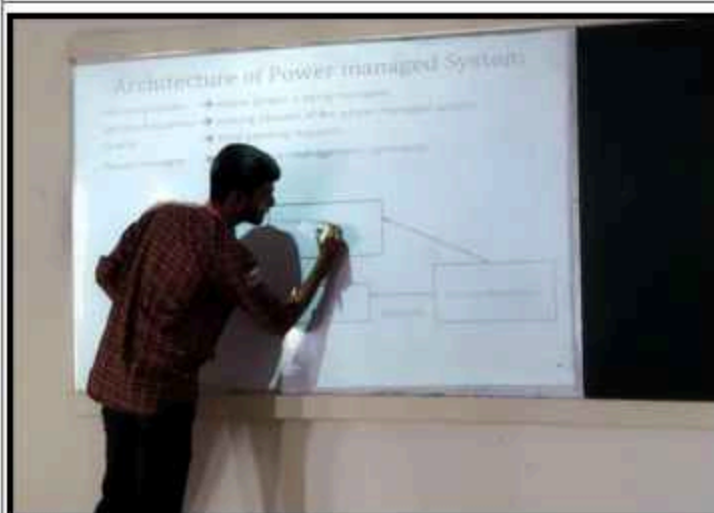
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Design Example: Model Train Controller - Design methodologies	Story Board	25.08.2021 (10 Minutes)
Specifications-System Analysis and Architecture Design	Mind Map	26.08.2021 (10 Minutes)
Platform-Level Performance Analysis	Open Book Test	03.09.2021 (15 Minutes)

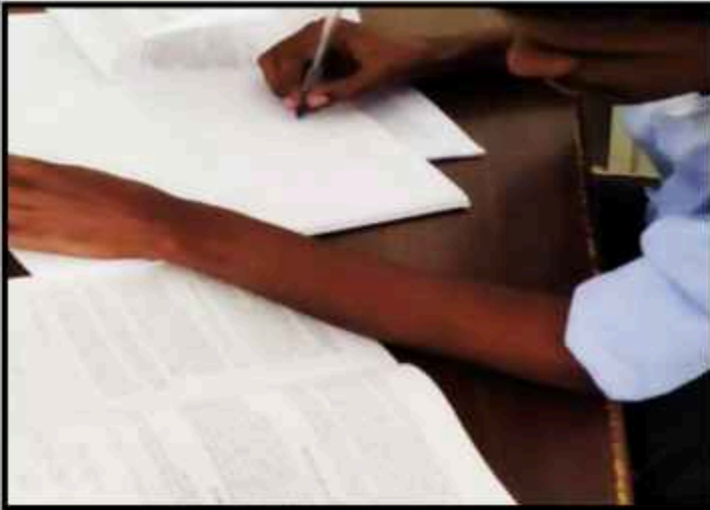
Story Board for Design Example: Model Train Controller - Design methodologies on 25.08.2021 (10 Minutes)



Mind Map for Specifications-System Analysis and Architecture Design on 26.08.2021 (10 Minutes)



**Open Book Test for Platform-Level Performance Analysis
on 03.09.2021 (15 Minutes)**





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Active Learning Description for Unit-II

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Course Code & Title: EC8791 Embedded and Real Time Systems

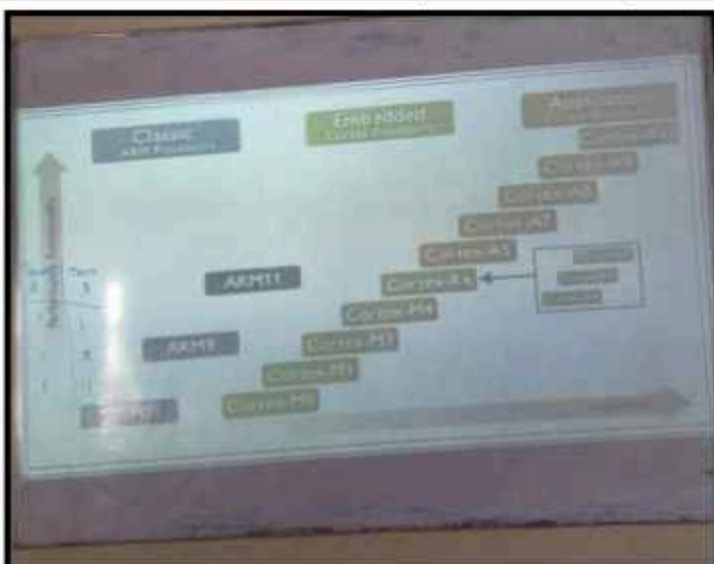
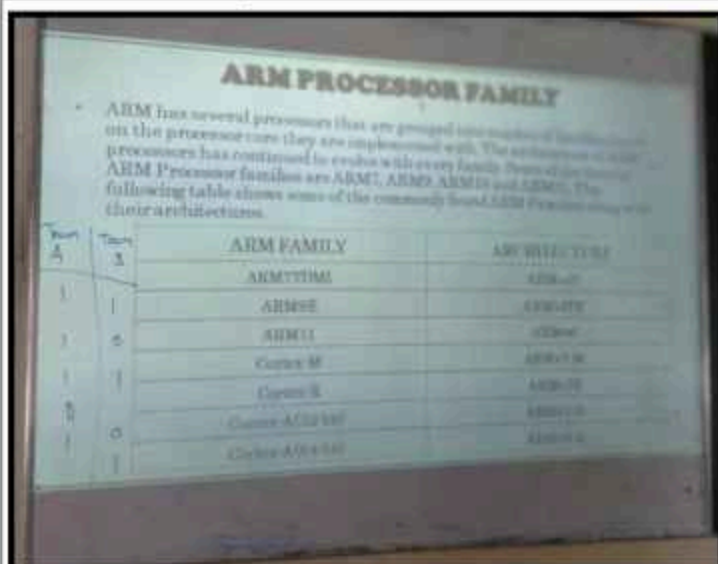
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Instruction Set of ARM LPC2148	Concept Tests	15.09.2021 (10 Minutes)
Pulse Width Modulation Unit	Visual Quiz	22.09.2021 (10 Minutes)
ARM Cortex M3 and MCU	In-Class Teams	24.09.2021 (15 Minutes)

Concept Tests for Instruction Set of ARM LPC2148 on 15.09.2021 (10 Minutes)



Visual Quiz for Pulse Width Modulation Unit on 22.09.2021 (10 Minutes)



In-Class Teams for ARM Cortex M3 and MCU on 24.09.2021 (15 Minutes)





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Active Learning Description for Unit-III

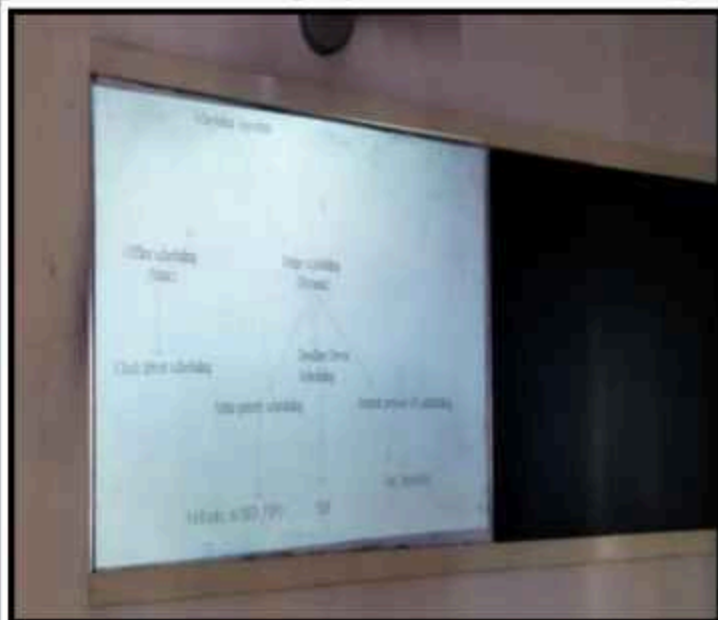
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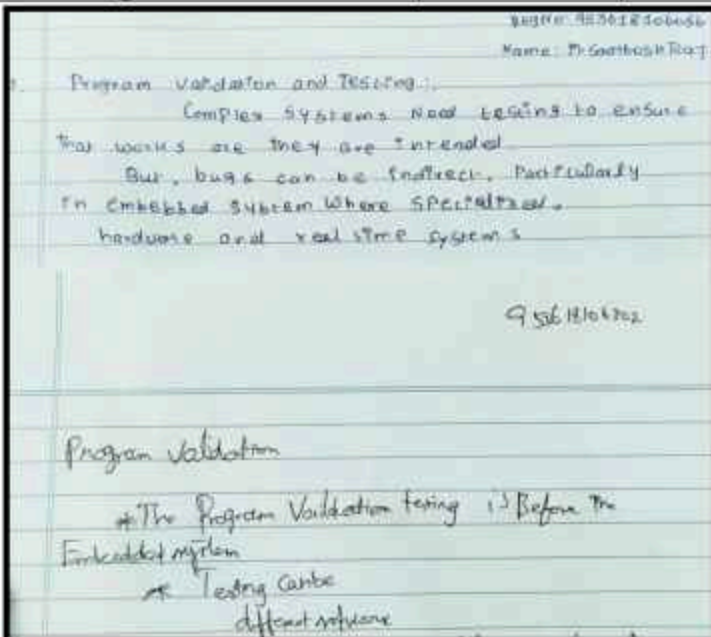
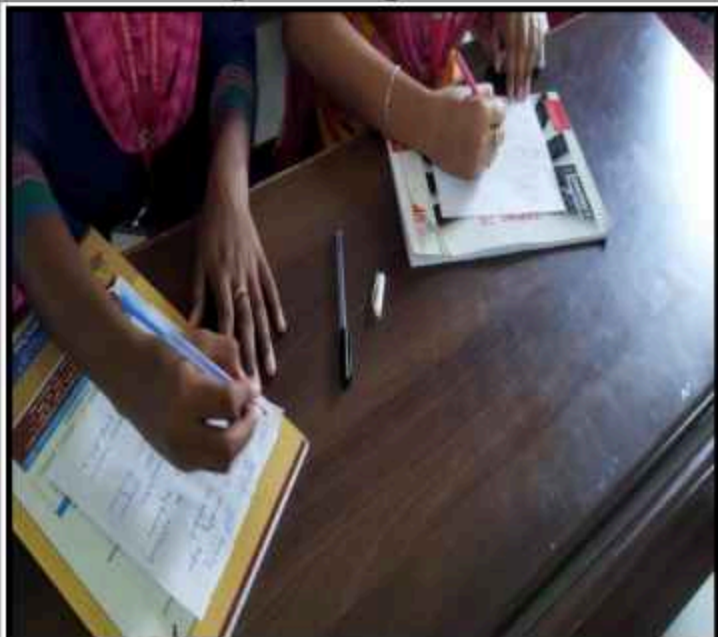
Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Models of Programs	Role-play	01.10.2021 (10 Minutes)
Program Validation and Testing	Exit Slip	20.10.2021 (10 Minutes)

Role-play for Models of Programs on 01.10.2021 (10 Minutes)



Exit Slip for Program Validation and Testing on 20.10.2021 (10 Minutes)





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Active Learning Description for Unit-IV

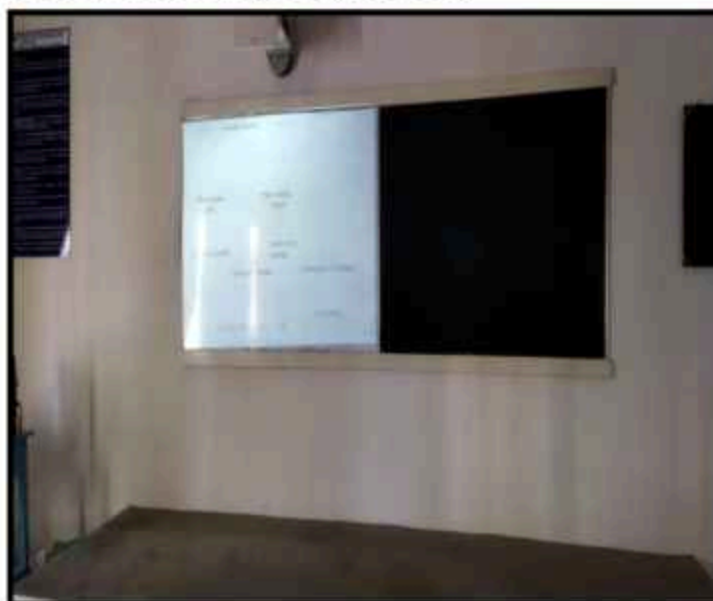
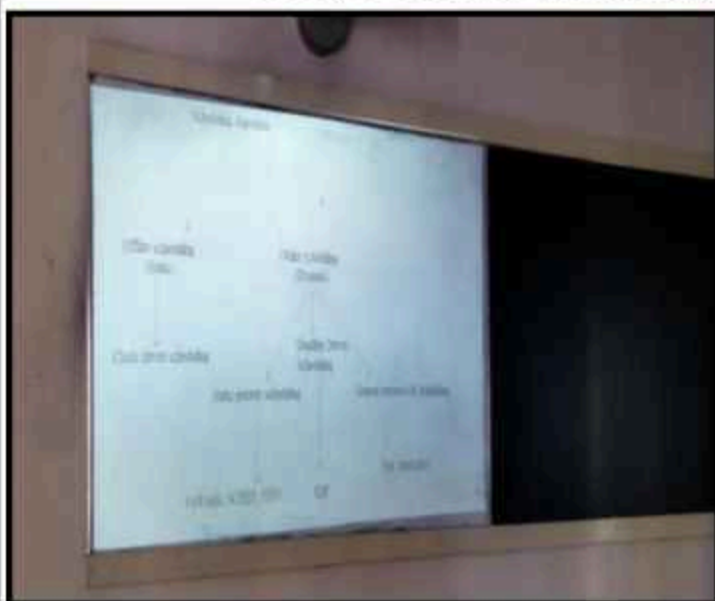
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Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Reliability	Story Board	09.11.2021 (10 Minutes)
Clock Synchronization	Open Book Test	12.11.2021 (15 Minutes)

Story Board for Reliability on 09.11.2021 (10 Minutes)



Open Book Test for Clock Synchronization on 12.11.2021 (15 Minutes)





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Active Learning Description for Unit-V

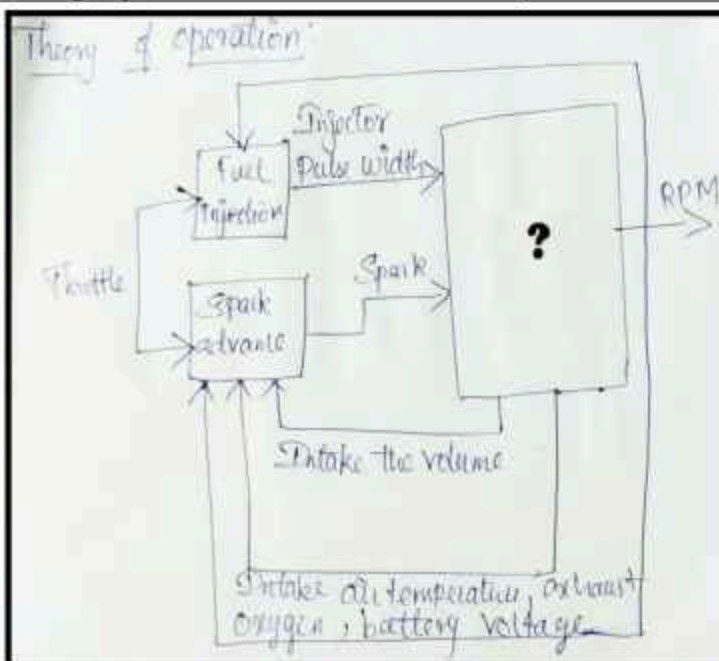
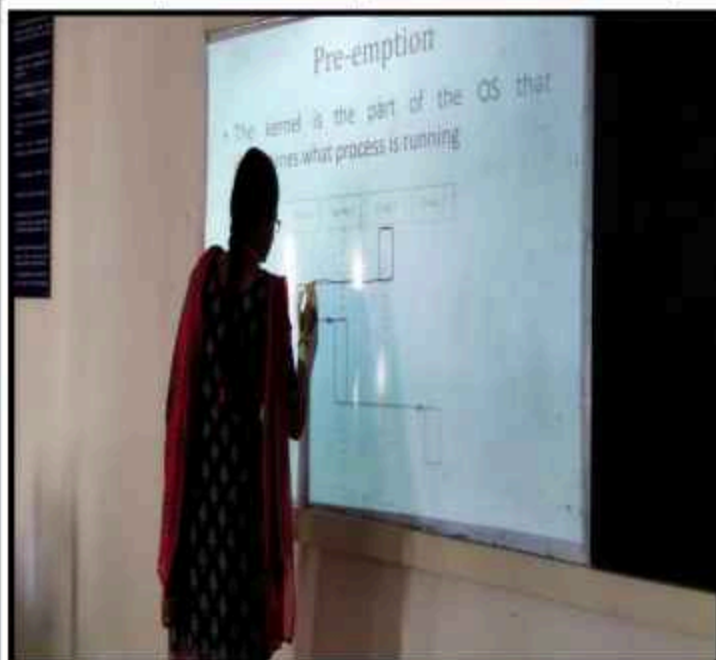
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Name of the Faculty member: Mr.B.Kannan

Name of the Topic	Name of the Innovative Practice	Date & Duration
Preemptive Real-Time Operating Systems	Mind Map	17.11.2021 (10 Minutes)
Audio Player, Engine Control Unit, Video Accelerator	Exit Slip	24.11.2021 (10 Minutes)

Mind Map for Preemptive Real-Time Operating Systems on 17.11.2021 (10 Minutes)



Exit Slip for Audio Player on 24.11.2021 (10 Minutes)

